

A No-Go Result for the Standard L^1 Decoupling Examples in the Type 2 Non-UMD Question

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Status

This is a candidate partial result. It does not solve the full question in Cox–Gorajski. It proves that two standard L^1 -based sources of non-UMD spaces with the decoupling property cannot produce a type 2 non-UMD example.

Source Metadata

The source paper is Sonja Cox and Mariusz Gorajski, *Vector-valued stochastic delay equations – a semigroup approach*, arXiv:1006.5349. The local source TeX line is `data/parsed/arxiv_sources/1006.5349/source.tex:541`. The rendered source evidence below is page 12 of the locally compiled source PDF.

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SONJA COX AND MARIUSZ GÓRAJSKI

4. THE STOCHASTIC DELAY EQUATION

4.1. The variation of constants formula. We now turn to the stochastic delay equation (SDE) as presented in the introduction and the related stochastic Cauchy problem (SDCP) on page 2. Recall that we assumed that (SDE) is set in a type 2 UMD Banach space E and that the related Cauchy problem is set in $\mathcal{E}^p(E) = E \times L^p(-1, 0; E)$ for some $p \in [1, \infty)$. (The results in this article are also valid if E is a type 2 Banach space with the decoupling property but we do not know of any such spaces that are not in fact UMD spaces.)

Recall that a Banach space F is said to have *type* $p \in [1, 2]$ if there exists a constant $C \geq 0$ such that for all finite choices $x_1, \dots, x_k \in F$ we have

$$\left(\mathbb{E} \left\| \sum_{j=1}^k \gamma_j x_j \right\|_F^2 \right)^{\frac{1}{2}} \leq C \left(\sum_{j=1}^k \|x_j\|_F^p \right)^{\frac{1}{p}},$$

where $(\gamma_j)_{j \geq 1}$ is a sequence of independent standard Gaussians. Hilbert spaces have type 2 and L^p -spaces with $p \in [1, \infty)$ have type $\min\{p, 2\}$. We refer to [1] for more information, for our purposes we only need that in Banach spaces with type 2 the following embedding holds, see p. 1460 in [19]:

$$(16) \quad L^2(0, t; \gamma(H, F)) \hookrightarrow \gamma(0, t; H, F).$$

Let $(\mathcal{T}(t))_{t \geq 0}$ denote the semigroup generated by $\mathcal{A}$, where $\mathcal{A}$ is the operator in (SDCP) defined by (1) in the introduction. We define the projections $\pi_1 : \mathcal{E}^p(E) \rightarrow E$ and $\pi_2 : \mathcal{E}^p(E) \rightarrow L^p(-1, 0; E)$ as follows:

$$\pi_1 \begin{bmatrix} x \\ f \end{bmatrix} = x; \quad \pi_2 \begin{bmatrix} x \\ f \end{bmatrix} = f.$$

The following property of $(\mathcal{T}(t))_{t \geq 0}$ is intuitively obvious and useful in the following:

$$(17) \quad \left(\pi_2 \mathcal{T}(t) \begin{bmatrix} x \\ f \end{bmatrix} \right)(u) = \pi_1 \mathcal{T}(t+u) \begin{bmatrix} x \\ f \end{bmatrix}.$$

for $f \in \mathcal{E}^p(E)$, $u \in [-1, 0]$, $t > -u$ (for a proof see [9], Proposition 3.11)

Source Question

In Section 4.1, Cox and Gorajski assume that their stochastic delay equation is set in a type 2 UMD Banach space E . They add the following parenthetical comment:

The results in this article are also valid if E is a type 2 Banach space with the decoupling property but we do not know of any such spaces that are not in fact UMD spaces.

Definitions and Background

A Banach space has Gaussian type 2 if there is a constant T such that for all finite choices $x_1, \dots, x_n$,

$$\left(\mathbb{E} \left\| \sum_{j=1}^n \gamma_j x_j \right\|^2 \right)^{1/2} \leq T \left(\sum_{j=1}^n \|x_j\|^2 \right)^{1/2},$$

where the γ_j 's are independent standard Gaussian variables.

We use the following standard facts.

1. Scalar $L^1(\mu)$ has cotype 2, and cotype 2 passes to subspaces and isomorphisms.
2. Kwapien's theorem says that a Banach space with type 2 and cotype 2 is isomorphic to a Hilbert space.
3. Hilbert spaces are UMD, and finite direct sums of UMD spaces are UMD.
4. Cox–Veraar proved that $L^1(S; Y)$ has the decoupling property when S is sigma-finite and Y is UMD.

Proof Intuition

The known non-UMD decoupling examples are L^1 -flavored. But type 2 is hostile to ℓ_1^n -geometry. In scalar L^1 subspaces the obstruction is even stronger: scalar L^1 has cotype 2, so adding type 2 collapses the space to Hilbert by Kwapien's theorem. In the Bochner examples $L^1(S; Y)$, an infinite or nonatomic measure algebra gives isometric copies of ℓ_1^n for arbitrarily large n , contradicting type 2. Thus the only type 2 Bochner cases are finite L^1 -sums of the original UMD space Y .

Main Partial Result

Theorem 1 (No-go for the standard L^1 decoupling examples). *Let E be a Banach space belonging to either of the following two classes.*

1. *E is isomorphic to a closed subspace of a scalar $L^1(\mu)$.*
2. *$E = L^1(S; Y)$, where (S, Σ, μ) is sigma-finite and Y is UMD.*

If E has type 2, then E is UMD. More precisely, in case (1), E is isomorphic to a Hilbert space. In case (2), either $Y = \{0\}$, or the measure algebra of S has only finitely many atoms and E is isomorphic to a finite ℓ_1 -sum of copies of Y .

Proof. Assume first that E is isomorphic to a closed subspace of a scalar $L^1(\mu)$. Since $L^1(\mu)$ has cotype 2 and cotype 2 passes to closed subspaces and to isomorphic copies, E has cotype 2. By hypothesis E also has type 2. Kwapien's theorem therefore implies that E is isomorphic to a Hilbert space. Hence E is UMD.

Now assume that $E = L^1(S; Y)$, where S is sigma-finite and Y is UMD. If $Y = \{0\}$, there is nothing to prove, so choose $y_0 \in Y$ with $\|y_0\| = 1$. Suppose that S contains n pairwise disjoint measurable sets $A_1, \dots, A_n$, each of finite positive measure. Then

$$(a_1, \dots, a_n) \mapsto \sum_{k=1}^n a_k \mu(A_k)^{-1} \mathbf{1}_{A_k} y_0$$

is an isometric embedding of ℓ_1^n into $L^1(S; Y)$. Indeed, the supports are disjoint, so the $L^1(S; Y)$ -norm of the image is $\sum_{k=1}^n |a_k|$.

If such disjoint families existed for arbitrarily large n , then $L^1(S; Y)$ would contain isometric copies of ℓ_1^n for arbitrarily large n . This is incompatible with type 2: applying the type 2 inequality inside ℓ_1^n to the coordinate vectors $e_1, \dots, e_n$ gives

$$\left(\mathbb{E} \left(\sum_{k=1}^n |\gamma_k| \right)^2 \right)^{1/2} \leq T\sqrt{n},$$

whereas the left side is at least $n \mathbb{E}|\gamma_1| = \sqrt{2/\pi} n$. Thus the type 2 constant would have to grow at least like $\sqrt{n}$, a contradiction.

Therefore a type 2 space $L^1(S; Y)$ admits a uniform bound on the size of a family of pairwise disjoint positive finite-measure sets. Since S is sigma-finite, this forces the measure algebra to be finite atomic: a nonatomic positive-measure part can be split into arbitrarily many positive finite measure pieces, and infinitely many atoms also give arbitrarily large disjoint families. Hence $L^1(S; Y)$ is isomorphic, indeed after rescaling atom weights isometric, to a finite ℓ_1 -direct sum of copies of Y . Finite sums preserve the UMD property, so E is UMD. $\square$

Consequence for the Cox–Gorajski Aside

The theorem rules out the most visible L^1 -based non-UMD decoupling routes as answers to the Cox–Gorajski parenthetical question. In particular, the non-UMD examples $L^1(S; Y)$ from Cox–Veraar cannot simultaneously have type 2 unless the measure part is finite atomic, in which case they are UMD. Likewise, scalar L^1 -subspace examples with type 2 are Hilbertian.

Limitations and Novelty Check

This is only a partial no-go result. It does not prove that every type 2 space with the decoupling property is UMD. It also does not address hypothetical decoupling examples outside the scalar L^1 -subspace and Cox–Veraar Bochner $L^1(S; Y)$ families.

The lightweight run indexes were searched for arXiv:1006.5349, the source title, and the terms `type 2`, `decoupling property`, `UMD`, `L1`, `Cox Veraar`, and related local source-corpus keywords. No duplicate packet for this exact partial no-go theorem was found.

References

References

- [1] S. Cox and M. Gorajski, *Vector-valued stochastic delay equations – a semigroup approach*, arXiv:1006.5349.
- [2] S. Cox and M. Veraar, *Some remarks on tangent martingale difference sequences in L^1 -spaces*, arXiv:0801.0695.
- [3] N. J. Kalton, E. Lorist, and L. Weis, *Euclidean structures and operator theory in Banach spaces*, arXiv:1912.09347. See the discussion of the Kwapien–Maurey theorem and the type 2 plus cotype 2 Hilbertian consequence.
- [4] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, second edition, Springer, 2016.